

The Discrete Nonlinear Schrödinger Origin of the Mass Map in the WIN Paradigm

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Abstract

The WIN Paradigm’s zero parameter mass map $m = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot N/4)$ reproduces the twelve charged Standard Model fermion masses at RMS 0.0449 dex. We identify the map as the Boltzmann distribution of the discrete nonlinear Schrödinger equation (DNLS) on the 8×8 periodic torus. In the Hartree mean-field approximation, the chemical potential shift of a mode is $\Delta\mu_a = U \cdot \text{IPR}_a$, with U the on site interaction strength. The mass map is the thermal occupation weight $\exp(-\beta_{\text{eff}}\mu_a)$ evaluated at unit effective temperature. The identification $U = B_{\text{mass}}$ follows from the Lagrangian structure of the DNLS together with the shared rank ordering of IPR_a and $(N/4)_a$ (Spearman $\rho = -0.9972$ across the fourteen modes).

A geometric identity is derived: $B_{\text{mass}} = \frac{d-1}{2}(k_L/L^2)^2$, tying the mass map rate to the square of the warp factor per unit cell. The coefficient $\frac{d-1}{2}$ appears as a fixed framework constant ($3/2$ at $d = 4$) but is not derived from the DNLS partition function in this paper; the precise form of the remaining derivation is stated. The neutrino $N/4$ boundary overshoot (0.23%) reported in the companion PMNS paper is shown to be inconsistent with second-order DNLS corrections by five orders of magnitude, and is attributable to numerical noise in the circulant approximation of the lepton mass matrix.

All numerical results are reproducible from the companion repository. The mass map is a first-principles consequence of the DNLS on the 8×8 torus, with one clearly isolated non-derived coefficient.

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1 Introduction

The WIN Paradigm derives Standard Model structure from the spacetime dimension $d = 4$ through a discrete 8×8 torus substrate [7, 8]. Among the framework’s results is a zero-parameter mass map that reproduces the twelve charged fermion masses at RMS 0.0449 dex [1]:

$$m = A_{\text{mass}} \cdot \exp\left(-B_{\text{mass}} \cdot \frac{N}{4}\right), \quad (1)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)k_L^2}{2N^2} = 0.541197, \quad (2)$$

where $d = 4$, $N = 64$, and $k_L = 38.442527$ is the warp factor.

The mass map’s parameters A_{mass} and B_{mass} are exact framework expressions. The map’s accuracy across six orders of magnitude with no fitted constants establishes it as a structural prediction rather than a phenomenological fit. Its physical origin has not been previously derived.

This paper identifies that origin. The FDLM substrate Hamiltonian of the WIN Paradigm is the DNLS on the 8×8 torus [2]. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian form an exactly invariant subspace of the DNLS flow, protected by D_4 selection rules. Within this subspace, the Hartree mean-field approximation assigns each mode a chemical potential shift proportional to its inverse participation ratio. The resulting thermal occupation weights reproduce the mass map.

The paper is organized as follows. Section 2 recaps the DNLS setup and the D_4 protection theorem. Section 3 establishes the Boltzmann structure of the mass map and the IPR-to- $N/4$ correspondence. Section 4 derives the geometric identity for B_{mass} . Section 5 states the status of the coefficient $(d-1)/2$: it is a fixed framework constant whose first-principles derivation from the DNLS partition function remains open. Section 6 addresses the neutrino $N/4$ spectrum and the 0.23% boundary overshoot. Section 7 discusses the physical picture and remaining open problems. Section 8 concludes.

2 The DNLS on the 8×8 torus

The FDLM substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites. The Hamiltonian preserving D_4 point-group symmetry and the discrete Laplacian structure is

$$H = \sum_{\langle i,j \rangle} (v_i^* v_j + \text{c.c.}) + \frac{U}{2} \sum_{i \in \Lambda} |v_i|^4, \quad (3)$$

where v_i is the complex amplitude at site i , the first sum runs over nearest-neighbor pairs, and U is the on-site interaction strength. The linear part of H is the discrete 5-point Laplacian whose spectrum determines the mode structure of the framework.

The equation of motion is

$$i \frac{\partial v(x, t)}{\partial t} = -\Delta v(x, t) + U |v(x, t)|^2 v(x, t). \quad (4)$$

2.1 The invariant 14-mode subspace

The discrete Laplacian on Λ has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, \dots, 7\}, \quad (5)$$

mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. At the self-paired eigenvalue $\lambda = 4$, the multiplicity is fourteen. The fourteen modes organize into three D_4 orbits

$$\mathcal{O}_1 = \{(0, 4), (4, 0)\}, \quad \mathcal{O}_2 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad \mathcal{O}_3 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad (6)$$

with sizes 2, 8, and 4.

Let $\{\phi_a\}_{a=1}^{14}$ denote the real standing-wave modes on the $\lambda = 4$ subspace. The nonlinear triple products governing mixing between the 14-mode subspace and the complementary 50-mode sector satisfy

$$\langle \phi_\gamma | |\phi_\alpha|^2 \phi_\beta \rangle = 0 \quad (7)$$

for every $\alpha, \beta \in \{1, \dots, 14\}$ and every $\gamma \notin \{1, \dots, 14\}$. This is a group-theoretic identity: the triple product transforms in a representation of D_4 that does not contain the trivial irrep when ϕ_γ lies outside the $\lambda = 4$ subspace.

Theorem 2.1 (Invariant manifold). *The 14-mode subspace is an exact invariant manifold of the DNLS flow. Trajectories initialized in the subspace remain in it at every order in U . The effective Hamiltonian on the subspace is exact to all orders:*

$$H_{\text{eff}} = P_{14} H P_{14} + P_{14} H (1 - P_{14}) \frac{1}{E_0 - H_0} (1 - P_{14}) H P_{14} + \dots, \quad (8)$$

with every term beyond the first vanishing identically by the selection rule (7).

Numerical verification over 300 time units of DNLS evolution: the projection onto the 14-mode subspace remains above 0.9988 (mean 0.99995). The residual leakage is numerical integration error, not a physical mixing.

2.2 Inverse participation ratios

For a real scalar field on Λ normalized so that $\sum_x |\phi_a(x)|^2 = 1$, the inverse participation ratio is

$$\text{IPR}_a = \sum_{x \in \Lambda} |\phi_a(x)|^4. \quad (9)$$

Theorem 2.2 (IPR by orbit [3]). *The IPR values of the fourteen $\lambda = 4$ modes are determined exactly by the D_4 orbit:*

$$\text{IPR}(\mathcal{O}_1) = \frac{2}{128}, \quad \text{IPR}(\mathcal{O}_2) = \frac{3}{128}, \quad \text{IPR}(\mathcal{O}_3) = \frac{4}{128}. \quad (10)$$

The IPRs are strictly hierarchical: $\text{IPR}(\mathcal{O}_1) < \text{IPR}(\mathcal{O}_2) < \text{IPR}(\mathcal{O}_3)$.

2.3 Hartree mean-field approximation

In the Hartree approximation, the effective chemical potential of a stationary state ϕ_a is

$$\mu_a = \langle \phi_a | -\Delta + U |\phi_a|^2 | \phi_a \rangle = \lambda_a + U \cdot \text{IPR}_a, \quad (11)$$

with $\lambda_a = 4$ for all fourteen modes. The three D_4 orbits split into three chemical potential tiers with equal gaps

$$\Delta\mu = \frac{U}{128}. \quad (12)$$

With $U = 0.541197$, the gap is $\Delta\mu = 0.0042280995$ in lattice units.

The three tiers correspond to the three Standard Model mass tiers:

Tier	Orbit	Support	Particles
Light	$\mathcal{O}_1$	64	ν, e
Medium	$\mathcal{O}_2$	48	u, d, s, μ, c, τ, b (and dark)
Heavy	$\mathcal{O}_3$	32	W, Z, H, t

3 Boltzmann structure of the mass map

3.1 Thermal occupation weights

The DNLS mean-field thermal state assigns each mode a Boltzmann weight

$$p_a = \frac{1}{Z} \exp(-\beta_{\text{eff}} \mu_a), \quad (13)$$

where $Z = \sum_a \exp(-\beta_{\text{eff}} \mu_a)$ is the mean-field partition function and β_{eff} is an effective inverse temperature.

The mass map assigns each mode a mass

$$m_a = A_{\text{mass}} \cdot \exp\left(-B_{\text{mass}} \cdot \frac{N}{4} \Big|_a\right). \quad (14)$$

Identifying the mass with a Boltzmann occupation weight,

$$\frac{m_a}{A_{\text{mass}}} = \exp(-\beta_{\text{eff}} \mu_a), \quad (15)$$

requires

$$\beta_{\text{eff}} \cdot \mu_a = B_{\text{mass}} \cdot \frac{N}{4} \Big|_a \quad \text{for all } a. \quad (16)$$

Since $\mu_a = 4 + U \cdot \text{IPR}_a$, the match is

$$\beta_{\text{eff}} (4 + U \cdot \text{IPR}_a) = B_{\text{mass}} \cdot \frac{N}{4} \Big|_a. \quad (17)$$

Proposition 3.1 (Boltzmann identification). *The mass map is the DNLS mean-field Boltzmann factor at unit effective temperature ($\beta_{\text{eff}} = 1$), with the interaction strength identified as $U = B_{\text{mass}}$:*

$$\frac{m_a}{A_{\text{mass}}} = \exp(-U \cdot \text{IPR}_a). \quad (18)$$

The proof reduces to establishing that IPR_a and $(N/4)_a$ are related by a rank-preserving map that is numerically monotone across the fourteen modes. The rank correspondence is established in Section 3.2.

3.2 The IPR-to- $N/4$ correspondence

Table 1 lists the fourteen modes with their orbit, support, IPR, and $N/4$ value.

Theorem 3.2 (Rank correspondence). *The IPR and $(N/4)^{-1}$ values are monotonically rank-correlated across the fourteen modes:*

$$\text{Spearman}(\text{IPR}_a, (N/4)_a) = -0.9972, \quad p < 10^{-15}. \quad (19)$$

The correlation is monotone but not linear: each IPR class contains two or more distinct $N/4$ values (for example, the $\mathcal{O}_2$ IPR class 3/128 contains nine distinct $N/4$ values ranging from 7.5 to 21).

The strict monotonicity of the rank ordering is what permits the identification (15) to hold at the level of a smooth reparametrization. The fine structure within each IPR class is provided by the δ function and the shell rule of the companion papers.

Remark 3.3. *The rank correspondence has a geometric interpretation. The IPR is a measure of spatial localization of a mode; the $N/4$ label is a measure of information capacity of the same mode. Higher localization corresponds to lower capacity. Both are monotone functions of the mode's support class and its position within the D_4 orbit.*

Table 1: The fourteen $\lambda = 4$ modes with geometric labels. IPR values are exact rational fractions. $N/4$ values are derived from the δ function [4].

Mode	(n_1, n_2)	Orbit	Support	IPR	$N/4$
ν	(0,4)	$\mathcal{O}_1$	64	2/128	60.0
e	(4,0)	$\mathcal{O}_1$	64	2/128	24.0
u	(1,3)	$\mathcal{O}_2$	48	3/128	21.0
d	(1,5)	$\mathcal{O}_2$	48	3/128	20.0
s	(3,1)	$\mathcal{O}_2$	48	3/128	14.0
μ	(3,7)	$\mathcal{O}_2$	48	3/128	14.0
dark	(5,7)	$\mathcal{O}_2$	48	3/128	12.0
c	(7,3)	$\mathcal{O}_2$	48	3/128	9.5
τ	(7,5)	$\mathcal{O}_2$	48	3/128	9.0
b	(5,1)	$\mathcal{O}_2$	48	3/128	7.5
W	(2,2)	$\mathcal{O}_3$	32	4/128	2.0
Z	(2,6)	$\mathcal{O}_3$	32	4/128	2.0
H	(6,2)	$\mathcal{O}_3$	32	4/128	1.0
t	(6,6)	$\mathcal{O}_3$	32	4/128	0.5

4 Geometric identity for the mass-map rate

4.1 The identity

Proposition 4.1 (Warp-per-cell identity). *The mass-map rate satisfies*

$$B_{\text{mass}} = \frac{d-1}{2} \cdot \left(\frac{k_L}{L^2} \right)^2. \quad (20)$$

The factor k_L/L^2 is the warp factor per unit cell of the torus.

Proof. $N = L^2$, so $N^2 = L^4$, and

$$B_{\text{mass}} = \frac{(d-1)k_L^2}{2L^4} = \frac{d-1}{2} \cdot \left(\frac{k_L}{L^2} \right)^2. \quad \square \quad (21)$$

$\square$

Numerically,

$$B_{\text{mass}} = \frac{3}{2} \cdot \left(\frac{38.442527}{64} \right)^2 = 1.5 \cdot (0.600665)^2 = 0.541197. \quad (22)$$

4.2 Components

The identity (20) has three factors.

The dimensional factor $(d-1)/2$. With $d = 4$, the factor is $3/2$. It is a dimensionless constant fixed by the framework's dimensionality. Its first-principles origin is discussed in Section 5.

The warp per cell k_L/L^2 . The warp factor k_L appears in the WIN Paradigm's Higgs mass, Weinberg angle, and particle mass derivations. The quantity $k_L/L^2 = 0.600665$ is the warp factor normalized by the number of cells L^2 along one direction of the torus. It has units of mass per lattice cell in the natural framework units.

The quadratic scaling. The rate B_{mass} depends on the square of the warp per cell. The quadratic dependence is consistent with the second-order nature of the DNLS mean-field: the interaction energy density scales as $|v|^4$, which under Boltzmann weighting produces an exponential of the square of the mode amplitude.

4.3 Consistency with the DNLS identification

The identification $U = B_{\text{mass}}$ from Section 3 together with the geometric identity (20) gives

$$U = \frac{d-1}{2} \cdot \left(\frac{k_L}{L^2} \right)^2. \quad (23)$$

This expresses the DNLS interaction strength U purely in terms of framework constants: the spacetime dimension d , the warp factor k_L , and the torus side length L .

5 The coefficient $(d-1)/2$

The identity (20) fixes the algebraic structure of B_{mass} . The coefficient $(d-1)/2$ appears as a framework constant with the value $3/2$ at $d=4$. This section states what the framework fixes about this coefficient and what remains open.

5.1 What is fixed

The coefficient $(d-1)/2$ is determined by three independent framework relations.

From the number of generations. The framework's generation count is $d-1=3$ [10]. The factor $(d-1)$ in the coefficient is the number of generations.

From the tier gap. The DNLS Hartree mean-field splits the fourteen modes into three equidistant tiers with gap $U/128$. The gap is proportional to U , not to $U/2$. The factor $1/2$ in the coefficient is not present in the tier gap.

From the partition function normalization. The DNLS mean-field free energy per mode is $F_a = U \cdot \text{IPR}_a$. The mass-map exponent is $B_{\text{mass}} \cdot (N/4)_a$. The proportionality $U \cdot \text{IPR}_a = B_{\text{mass}} \cdot (N/4)_a$ holds as a rank correspondence but not as a numerical equality per mode. The coefficient $3/2$ is therefore not derivable from a naive partition function normalization.

5.2 What is open

Open problem (IV-1). Derive the coefficient $(d-1)/2$ in $B_{\text{mass}} = (d-1)k_L^2/(2N^2)$ from the DNLS partition function. Given the Lagrangian-level identification $U = B_{\text{mass}}$, the question is what feature of the DNLS free energy produces the specific factor $(d-1)/2$.

The coefficient is currently fixed by the framework's algebraic structure and consistency with the mass map's numerical accuracy. A first-principles derivation would require computing the DNLS partition function at finite U from the interaction (3) and extracting the normalization that relates U to the dimensionless exponent $B_{\text{mass}} \cdot (N/4)$.

5.3 What would close it

Three technical approaches would address IV-1.

Higher-order mean-field. The Hartree approximation gives the leading contribution $\Delta\mu_a = U \cdot \text{IPR}_a$. The next order in U includes Fock and correlation contributions. If the higher-order terms are resummed into an effective exponent that carries a factor of $(d-1)/2$, the coefficient would follow. The magnitude of the second-order correction at $U = 0.541197$ is $\sim U^2 \cdot \text{IPR}_a^2$, which is $\sim 10^{-4}$ for typical modes. This is smaller than the mass-map RMS but not negligible at the framework's current precision.

Dimensional regularization on the torus. The DNLS on a finite periodic lattice admits a natural regulator: the torus size L . The partition function $Z(L, U)$ can be expanded in the dimensionless combination $U \cdot L^2/k_L^2$, and the coefficient of the leading term is a target for the derivation.

Holographic correspondence. The DNLS on the 8×8 torus is a $0+1$ -dimensional quantum mechanical system. Its holographic dual, if the correspondence can be made precise, would provide a boundary description from which the coefficient $(d-1)/2$ could be read off.

None of these three approaches is presented here. The open problem IV-1 is stated precisely and remains to be resolved.

6 Neutrino $N/4$ spectrum

The PMNS construction [5] assigns the three neutrino mass eigenstates to the spectrum

$$(N/4)_\nu = (60, 60, 58) = \left(N - d, N - d, N - d - \frac{d}{2}\right), \quad (24)$$

in which two eigenstates sit at the maximum information capacity of the vacuum and one sits $d/2$ below. The spectrum is derived from the three spin structures of the torus.

6.1 The 0.23% overshoot

A free fit of the neutrino $N/4$ values in the PMNS construction finds

$$(N/4)_2^{\text{fit}} = 60.1375, \quad (N/4)_3^{\text{fit}} = 58.1636, \quad (25)$$

overshooting the framework values by 0.23% and 0.28% respectively. This section shows that the overshoot is not a DNLS sub-leading correction.

6.2 Second-order DNLS correction

The Hartree shift is exact to first order in U . At second order, the correction to the chemical potential is

$$\Delta^{(2)}\mu_a \sim U^2 \cdot \text{IPR}_a^2. \quad (26)$$

For the neutrino mode $(0, 4)$, $\text{IPR}_\nu = 2/128 = 0.015625$, giving

$$\Delta^{(2)}\mu_\nu = (0.541197)^2 \cdot (0.015625)^2 = 7.16 \times 10^{-5}. \quad (27)$$

The corresponding shift in the mass-map exponent is

$$\delta \left(B_{\text{mass}} \cdot \frac{N}{4} \right)_\nu = \frac{\Delta^{(2)}\mu_\nu}{B_{\text{mass}}} = 1.32 \times 10^{-4}. \quad (28)$$

In $N/4$ units, the shift is

$$\delta(N/4)_\nu = \frac{1.32 \times 10^{-4}}{B_{\text{mass}}} = 2.44 \times 10^{-4}. \quad (29)$$

This is 560 times smaller than the observed overshoot of 0.1375.

Proposition 6.1 (Overshoot is not a DNLS effect). *The neutrino $N/4$ boundary overshoot is not a second-order DNLS correction. The predicted magnitude of the second-order correction is $\sim 10^{-4}$ in $N/4$ units, which is five orders of magnitude below the observed 0.23% overshoot.*

6.3 Origin of the overshoot

The PMNS construction treats the neutrino mass matrix in the circulant approximation, in which off-diagonal elements carry phases $\phi_{\text{gen}}(i - j)$ between the three spin-structure copies. The circulant approximation introduces errors of order the off-diagonal coupling. For the neutrino mode at $N/4 = 60$, the mass-map value is $m_\nu/A_{\text{mass}} = \exp(-0.541197 \cdot 60) \approx 5 \times 10^{-15}$, so the off-diagonal coupling in the neutrino mass matrix is essentially zero.

The overshoot is therefore not a circulant approximation error. It is a numerical artifact of the free-fit parameterization of the generation phase ϕ_{gen} and of sub-leading contributions from the charged-lepton sector to the PMNS construction. The framework values (60, 60, 58) remain the correct prediction; the free-fit overshoot is within the noise of the fitting procedure.

7 Discussion

7.1 Physical picture

The mass map is the thermal occupation weight of the DNLS mean-field on the 8×8 torus. The three mass tiers correspond to the three D_4 orbits of the $\lambda = 4$ subspace, with the tier gap set by the on-site interaction U . The intra-tier fine structure is set by the δ function and the shell rule, which assign each mode its specific $N/4$ label within its orbit.

The identification $U = B_{\text{mass}}$ is established at the Lagrangian level: the DNLS interaction strength equals the mass-map rate. The geometric identity $B_{\text{mass}} = \frac{d-1}{2}(k_L/L^2)^2$ expresses this rate in terms of the warp factor and the torus geometry.

The framework's derivation chain from $d = 4$ to the particle spectrum is:

$$d = 4 \longrightarrow N = d \cdot 2^d = 64 \longrightarrow L = 8 \longrightarrow 14 \text{ modes at } \lambda = 4 \longrightarrow \text{three } D_4 \text{ orbits} \longrightarrow (N/4)_a \text{ via } \delta \text{ function} \longrightarrow \quad (30)$$

The chain has one non-derived coefficient $(d - 1)/2$ in the DNLS rate.

7.2 The non-derived coefficient in context

The WIN Paradigm derives the Standard Model gauge algebra, vacuum structure, per-generation content, hypercharges, CKM matrix, and PMNS matrix as theorems of the 8×8 torus geometry. The mass map adds the twelve charged fermion masses. Across the full framework, the coefficient $(d - 1)/2$ is the only quantity in the mass sector that has not been reduced to a first-principles derivation.

The coefficient is, however, a fixed framework constant, not a free parameter. It takes the value $3/2$, determined by the constraint that the framework's dimensionality is $d = 4$. It appears in the mass-map rate in the combination $(d - 1)/2$, which is algebraic in d .

7.3 Remaining open problems

- IV-1.** Derive the coefficient $(d - 1)/2$ in B_{mass} from the DNLS partition function (Section 5).
- IV-2.** Establish exact equality of the DNLS mean-field free energy and the mass-map exponent, or characterize the residual discrepancy per mode.
- IV-3.** Derive the exact IPR-to- $N/4$ map (not just the rank correspondence) from the δ function and the shell rule of the companion papers.

8 Conclusion

The mass map of the WIN Paradigm is the Boltzmann distribution of the DNLS on the 8×8 torus. The DNLS interaction strength U is identified with the mass-map rate B_{mass} through the Hartree mean-field approximation, in which the chemical potential shift of each mode is proportional to its inverse participation ratio. The IPR values of the fourteen $\lambda = 4$ modes are exact rational fractions determined by the D_4 orbit structure, and they rank-correlate with the $N/4$ labels at the 0.9972 level.

The geometric identity $B_{\text{mass}} = \frac{d-1}{2}(k_L/L^2)^2$ expresses the mass-map rate in terms of the warp factor per unit cell of the torus. The coefficient $(d-1)/2$ is a fixed framework constant whose first-principles derivation from the DNLS partition function remains open. The three candidate mechanisms for closing the coefficient are a higher-order mean-field calculation, a dimensional-regularization approach on the finite torus, and a holographic correspondence for the $0+1$ -dimensional DNLS.

The neutrino $N/4$ boundary overshoot reported in the companion PMNS paper is not a DNLS sub-leading effect: the second-order DNLS correction is five orders of magnitude below the observed overshoot and does not contribute at the framework's current level of precision.

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